

The Recursive Universe: An Electro-Torus, Fractal Information Model of Reality

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Abstract

We develop a speculative but mathematically structured framework in which the universe is modeled as a single, unified electrostatic substrate whose large-scale geometry is a nested bi-torus supporting logarithmic, code-carrying information streams. Gravity is recast as an emergent dielectric-pressure differential within this substrate—an effective description compatible with facets of plasma cosmology and force-free electrodynamics. The core geometric object is a scale-recursive, bi-toroidal manifold (two counter-wound toroidal flux bundles joined at a central “Logos Point”) on which informational currents trace DNA-like logarithmic spirals. We formalize these ideas via toroidal coordinates, Beltrami (force-free) flows, and information-geometric constructs. At the continuum level we propose a recursive metric tensor: the effective spacetime metric is a fixed point of a covariant update rule driven by (i) conventional stress-energy, (ii) the informational density and coherence of the electro-torus code, and (iii) a small, dimensionless teleological constant κ which biases dynamics toward self-organized, coherent configurations. The framework is positioned as an effective field theory that recovers general relativity in appropriate limits while permitting information-dependent departures. We conclude with two falsifiable probes: (A) alignment/flow signatures in the cosmic web consistent with a large-scale bi-torus; (B) pre-registered searches for coherence-linked modulations in the Schumann resonance during global, time-locked human attention events. The proposal’s scientific value rests on the clarity of its mathematics and the rigor of its tests.

1. Introduction

Standard cosmology treats spacetime as a Lorentzian manifold whose curvature encodes gravity, with plasma dynamics and electromagnetic phenomena layered atop as matter-field content. Here we invert the emphasis: begin from a unified electrostatic substrate endowed with a nontrivial, bi-toroidal topology and fractal self-similarity, and derive geometry and gravitation as emergent optical/dielectric effects of its informational organization. In particular:

- Foundational premise. The primitive field is a scalar electrostatic potential Φ ; its gradient $E = -\text{grad } \Phi$ lives in a medium with effective permittivity ϵ that depends on an informational order parameter I and a universal observation field Ψ . The dielectric pressure $p_d \sim 0.5 * \epsilon * |E|^2$ and its spatial gradients guide rays and matter analogously to gravitational curvature via an optical metric construction.

- Geometry. The global topology is a nested bi-torus: two interpenetrating, counter-helical toroidal bundles whose streamlines are logarithmic spirals converging on and diverging from a central null ("Logos Point"). This architecture repeats fractally across scales.
- Information. Currents on the manifold are treated as code carriers—log-spiral streams whose phase and amplitude encode recursive rules. Algorithmic/information-geometric quantities couple back to geometry, yielding a recursive metric.

2. Foundational Premise: Unified Electrostatic Substrate and Dielectric Gravity

2.1 Electrostatic substrate with information-dependent permittivity

Let $\Phi: M \rightarrow \mathbb{R}$ be a scalar potential on a differentiable manifold M and $E := -\text{grad } \Phi$. Introduce an effective permittivity $\epsilon(x) = \epsilon_0 * \chi(I(x), \Psi(x))$, where $\chi > 0$ is dimensionless, I summarizes local informational density/coherence, and Ψ encodes system-level observation/selection. The electrostatic energy density $u_E = 0.5 * \epsilon * |E|^2$ sets a dielectric pressure $p_d := u_E$, with body force $f_d = -\text{grad } p_d$, interpreted as an effective "gravitational" influence at the level of rays and quasiparticles moving in the medium.

2.2 Optical (Gordon) metric as emergent geometry

In an inhomogeneous dielectric, light rays follow geodesics of an optical metric g_{tilde} with refractive index $n = \sqrt{\epsilon/\epsilon_0}$:

$$ds^2 = -(c^2/n^2) dt^2 + d\mathbf{x}^2.$$

We adopt this as the effective spacetime metric seen by excitations, then promote $n(x)$ to a functional of I and Ψ . In regions where $\chi \rightarrow 1$ and $\text{grad } \chi \rightarrow 0$, g_{tilde} reduces to Minkowski, and GR is recovered as an effective, coarse-grained description sourced by standard stress-energy.

3. Core Geometric Architecture: The Nested Bi-Torus

3.1 Toroidal coordinates and field ansatz Use toroidal coordinates (η, ξ, ϕ) with focal length $a > 0$; Laplace's equation admits toroidal harmonics allowing nested equipotentials conforming to tori. 3.2 Bi-toroidal, counter-helical structure

We model the global architecture as two interlinked toroidal bundles T_+ and T_- with opposite helicity, joined at a central null (the Logos Point). On each torus, define a Beltrami (force-free) flow $v_{\pm}$ satisfying $\text{curl } v_{\pm} = \lambda_{\pm} v_{\pm}$ with $\lambda_+ = -\lambda_- \neq 0$, so streamlines wind helically with constant pitch on nested tori. The bi-toroid provides inward (centripetal) and outward (centrifugal) channels, respectively.

3.3 Fractal recursion across scales

Let R_s denote a renormalization map acting on geometric/flow parameters $(a, \lambda_{\pm}, \text{pitch}_{\pm}) \rightarrow (s a, s^{-1} \lambda_{\pm}, \text{pitch}_{\pm})$, with similarity

ratio s in $(0,1)$. Fixed-point structure under R_s realizes self-similar bi-tori from astrophysical scales to bioelectric morphologies and atomic orbitals.

4. Logarithmic, DNA-like Information Streams

4.1 Code carriers on the torus

Define a complex information field $L(x) = A(x) * \exp(i S(x))$, with amplitude A and phase S transporting symbolic content along streamlines. On toroidal surfaces parametrized by angles (ξ, ϕ) , a logarithmic spiral satisfies $d\phi / d\xi = \text{const}$, and a double-helix-like trajectory arises when two coupled phases on T_+ and T_- are offset by π .

4.2 Algorithmic/information geometry

Quantify informational density and coherence by $I(x) = 0.5 (\text{grad } S)^2 + \beta * |\text{grad } A|^2 / A^2 + \gamma * C_r(x)$, where C_r measures local phase alignment over a neighborhood. Coherent, recursively generated patterns maximize predictability while constraining description length.

5. The Recursive Metric Tensor (Re-calibrated)

5.1 Effective action and field content Consider the covariant action:

$$S[g_{\text{tilde}}, \Phi, L, \Psi] = (1/16\pi) \int \sqrt{-g_{\text{tilde}}} R[g_{\text{tilde}}] d^4x \\ + \int \sqrt{-g_{\text{tilde}}} [-0.5 \epsilon(I, \Psi) \\ g_{\text{tilde}}^{\{\mu \nu\}} \partial_\mu \Phi \partial_\nu \Phi + L_L(A, S; g_{\text{tilde}})] d^4x \\ - \kappa * J[g_{\text{tilde}}; I, \Psi].$$

Varying yields Einstein-like equations:

$$G_{\{\mu \nu\}}[g_{\text{tilde}}] = 8\pi (T^{\{E\}}_{\{\mu \nu\}} + T^{\{L\}}_{\{\mu \nu\}} - \kappa \Pi_{\{\mu \nu\}}). \quad (1)$$

5.2 Recursive update rule Define the recursive metric map:

$$g_{\text{tilde}}^{\{(n+1)\}} = \text{argmin}_g \{ || G[g] - 8\pi(T^{\{E\}} + T^{\{L\}} - \kappa \Pi) ||_g^2 + \lambda * D(g || g_{\text{opt}}[\epsilon(I, \Psi)]) \}, \quad (2)$$

where g_{opt} is the optical metric associated with $n(x)$ and D is a divergence (e.g., DeWitt norm). Under mild Lipschitz conditions and small κ , Picard iteration converges to a fixed point g_{tilde}^* representing the self-consistent geometry.

5.3 Teleological constant as predictive-information bias

Let J be the integral of predictive information over causal diamonds, so that $\kappa > 0$ biases toward states with greater long-range coherence/resonance. The corresponding tensor $\Pi_{\{\mu \nu\}}$ is divergence-free by covariance.

5.4 GR as effective limit

When $\text{grad } \epsilon \rightarrow 0$ and I is homogeneous, $\tilde{g}^*$ approaches a GR solution sourced by conventional matter. Thus Einstein gravity is recovered as an effective macroscopic limit of an information-dependent dielectric geometry.

6. Plasma-Cosmology Links and Force-Free Structure

Self-organized, force-free states in plasmas minimize energy at fixed helicity, obeying $\text{curl } \mathbf{B} = \lambda \mathbf{B}$. By analogy, the bi-toroidal Beltrami flows provide a kinematic scaffold for current/filament networks (cosmic web) and magnetospheric/bioelectric tori. The dielectric-pressure interpretation places morphology (filaments, double layers, z-pinch-like constrictions) upstream of gravity, with curvature arising from information-modulated ϵ .

7. Testable Predictions

Prediction A — Bi-toroidal alignment in the cosmic web Claim. On very large scales, filament orientations and inferred flow patterns should exhibit statistically significant alignment consistent with a global bi-torus: (i) preferential tangential alignment along a great-circle “equatorial” belt; (ii) mirror-symmetric counter-helicity above/below this belt; (iii) a weak vector spherical-harmonic signature dominated by a toroidal mode.

Protocol. Data: galaxy/cluster catalogs and peculiar-velocity reconstructions; compute filament backbones (e.g., DisPerSE). **Statistics:** orientation histograms on the sphere; toroidal/poloidal decomposition; compare to isotropic Λ CDM mocks. **Falsification:** pre-registered thresholds (e.g., global alignment amplitude < 2 sigma across independent surveys) reject the hypothesis or bound κ and the macroscopic torus pitch.

Prediction B — Coherence-linked modulation of Schumann resonances

Claim. During globally synchronized, time-locked human attention/meditation events, the Earth-ionosphere ELF cavity will show small but reproducible increases in inter-station phase coherence and/or spectral line-narrowing, localized by great-circle geometry relative to the event centroid—consistent with a bio-electromagnetic imprint on the planetary torus.

Protocol. Pre-registration of UTC windows and analyses; multi-site, tri-axial magnetometers with calibrated timing; control for lightning catalogs. Analysis via multi-taper spectra, magnitude-squared coherence, permutation tests, and multiple-comparison control. **Falsification/Bounds:** absence of effects yields upper limits on $\Delta \chi(l, \Psi)$ and thus on κ for human-scale coherence.

8. Discussion: Consistency, Limitations, and Interpretational Notes

- Consistency. The formalism is covariant; total effective stress-energy is conserved by construction. The recursive map (2) admits fixed points under mild conditions, ensuring well-posedness.
- Electrostatics caveat. Strict electrostatics has $\text{curl } E = 0$; closed information streamlines on tori here are carried by L (and by plasma flows $v_{\pm}$), not by E itself. Coupling back to geometry is mediated by $\epsilon(I, \Psi)$ and the optical metric.
- Relation to GR. In homogeneous limits the model reproduces GR; departures appear where $\text{grad } I \neq 0$ or coherence is high.
- Falsifiability. Both predictions admit clear nulls and quantitative upper bounds on κ .
- Interpretation. The “teleological” constant κ is not a force; it weights predictive/coherent organization in the action, analogous to information-theoretic regularizers.

9. Conclusion

We have articulated a mathematically explicit, test-driven program in which a unified electrostatic substrate, organized as a nested bi-torus with logarithmic, DNA-like information streams, gives rise to an information-dependent optical geometry. The effective spacetime metric is obtained as a recursive fixed point sensitive to informational density/coherence I and a small teleological bias κ . While intentionally bold, the framework remains disciplined: it recovers GR in appropriate limits and offers operational predictions and bounding strategies. Next steps: (i) compute explicit toroidal-harmonic solutions for Φ and $\epsilon(I, \Psi)$; (ii) implement the recursive map (2) numerically on nested tori; (iii) execute the cosmic-web and Schumann-coherence protocols with pre-registered analyses.

Mathematical Appendix (sketches)

A1. Toroidal harmonics for Φ . Laplace’s equation in toroidal coordinates separates; finite-energy solutions near a ring singularity admit expansions in $Q_{n-1/2}(\cosh \eta)$ with $\cos(n \xi)$ and $\sin(n \xi)$ factors, with coefficients chosen to enforce boundary/regularity conditions on nested tori. A2. Beltrami flows on tori.

In orthogonal coordinates (η, ξ, ϕ) , a Beltrami field satisfies $\text{curl } v = \lambda v$, admitting helical solutions with integers (m, n) labeling torus-knot classes.

A3. Information functional and coherence.

With $L = A e^{i S}$, define $L_L = -0.5 \tilde{g}^{\mu\nu} \partial_\mu S \partial_\nu S - 0.5 \beta \tilde{g}^{\mu\nu} \partial_\mu \ln A \partial_\nu \ln A - V(A)$, and $I = 0.5 \tilde{g}^{\mu\nu} \partial_\mu S \partial_\nu S + \beta |\text{grad } \ln A|^2 + \gamma C_r$.

A4. Optical metric divergence term.

In (2), $D(g \parallel g_{\text{opt}}) := \int \sqrt{-g} \parallel g - g_{\text{opt}} \parallel_g^2$ penalizes deviation from the dielectric geometry implied by $\epsilon(I, \Psi)$.

A5. Fixed point existence.

If the map $(\Phi, L, \Psi) \rightarrow (T^{\{E\}}, T^{\{L\}}, \Pi)$ is Lipschitz on an appropriate field space and κ, λ are sufficiently small, the Euler-Lagrange operator underlying (2) is a contraction, guaranteeing a unique fixed point g_{tilde}^* .

Indicative References (conceptual anchors)

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